

Finding the right papers. Recovering the right information.

From scholarly discovery to machine-readable, evidence-linked scientific data



One research question hides two different problems

We will use the same example throughout the lecture.

How do additives and annealing conditions relate to operational stability in perovskite solar cells?

MINING FOR PAPERS

Which papers could answer it?

Search • rank • screen

MINING FROM PAPERS

What did each paper actually report?

Extract • normalize • link evidence

A complete literature-mining workflow needs both.

What this lecture will cover

1

Find papers systematically

Choose appropriate databases or APIs, combine search methods, screen, and document the evidence set.

2

Turn documents into usable content

Distinguish publisher XML or HTML, native PDF text, layout parsing, OCR, tables, figures, and supplements.

3

Extract and verify scientific fields

Use rules, NLP, supervised models, and LLMs where they fit then preserve context, evidence, and uncertainty.

The two jobs have different inputs, outputs, and success criteria

MINING FOR PAPERS

INPUT

Research question + search vocabulary

PROCESS

Search • rank • deduplicate • screen

OUTPUT

Documented set of relevant papers

SUCCESS: relevant studies are not missed
unnecessarily

MINING FROM PAPERS

INPUT

Included papers + extraction schema

PROCESS

Parse • extract • normalize • verify

OUTPUT

Evidence-linked scientific records

SUCCESS: values are correct, contextualized, and
traceable

Quick classification: mining FOR papers or mining FROM papers?

Decide what the desired output is.

1

Find recent papers on operational stability

FOR or THE?

2

Collect each device's annealing temperature

FOR or THE?

3

Identify papers comparing two transport layers

FOR or THE?

4

Extract the direction of a performance change

FOR or THE?

The desired output determines the task

A comparison can still be a retrieval task if the output is a set of papers.

1

**Find recent papers on
operational stability**

FOR

2

**Collect each device's annealing
temperature**

THE

3

**Identify papers comparing two
transport layers**

FOR

4

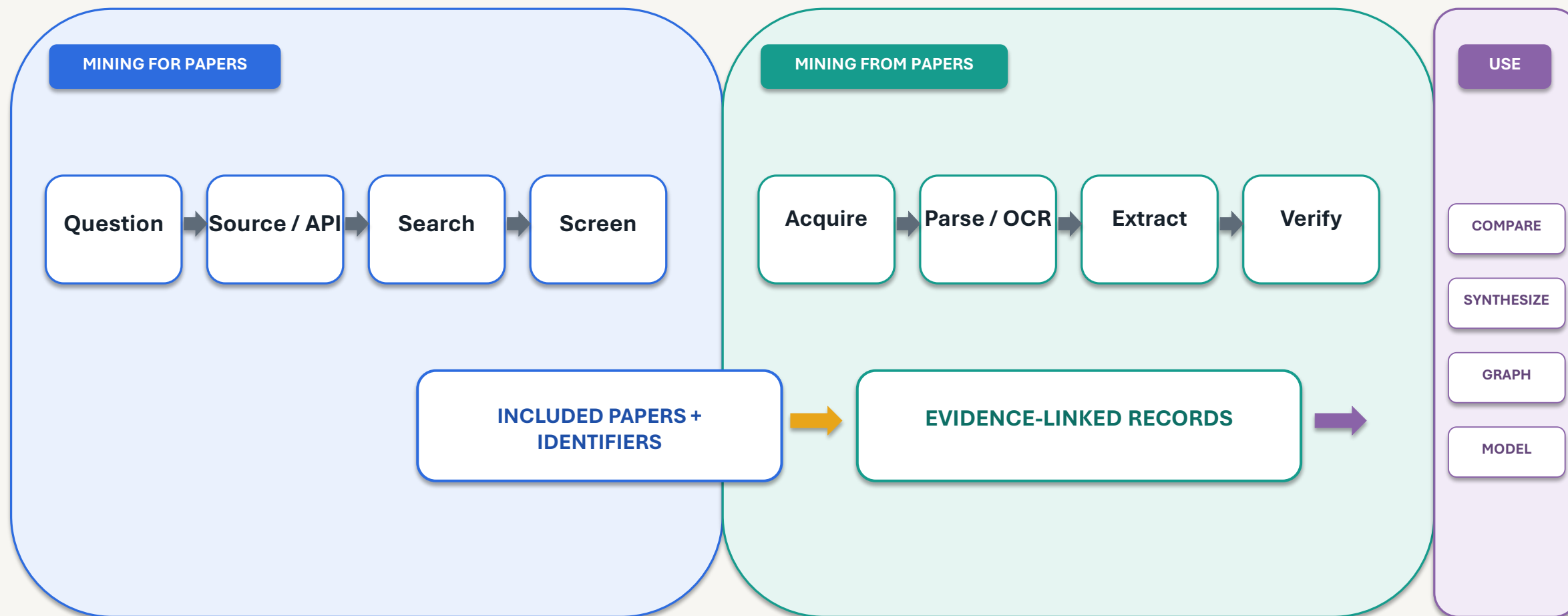
**Extract the direction of a
performance change**

THE

Task 3 asks for papers. Task 4 asks for a relationship inside the papers.

The complete workflow has two distinct transformations

First build a defensible paper set. Then convert the documents into evidence-linked scientific records.



Retrieval returns documents. Parsing recovers their content. Information extraction maps that content into scientific fields.

Mining for papers

Turn a scientific question into a documented, defensible evidence set.



Start with the scientific question — not with a pile of keywords

Break the question into concepts that must be represented in the search.

How do additives and annealing conditions relate to operational stability in perovskite solar cells?

SYSTEM

**perovskite solar
cells**

INTERVENTION

**additives +
annealing**

OUTCOME

**operational
stability**

CONTEXT

**test conditions /
device type**

Concept blocks help you add synonyms without losing the logic of the question.

Build a vocabulary for each concept

Authors may describe the same scientific idea with different words.

CONCEPT	SEED TERMS	ALTERNATE WORDING
SYSTEM	perovskite solar cell	metal-halide perovskite • halide perovskite photovoltaic
ADDITIVE FAMILY	additive • passivation	dopant • surface treatment • defect passivation
ANNEALING FAMILY	anneal • heat treatment	thermal processing • heated • post-deposition treatment
OUTCOME	operational stability	degradation • lifetime • retained efficiency

Different intervention families may deserve separate searches rather than one overloaded query.

Turn the concept blocks into transparent search expressions

Two intervention families can be searched separately, then merged and deduplicated.

QUERY A • ADDITIVES

("perovskite solar cell*" OR "metal-halide perovskite*")

AND

(additive* OR passivat* OR dop*)

AND

("operational stability" OR degrad* OR lifetime)

QUERY B • ANNEALING

("perovskite solar cell*" OR "metal-halide perovskite*")

AND

(anneal* OR "heat treatment" OR "thermal process*")

AND

("operational stability" OR degrad* OR lifetime)



MERGE RESULTS → DEDUPLICATE → SCREEN

Where do candidate papers come from?

Different platforms expose different layers of the scholarly record.

DOI + METADATA REGISTRY

Crossref

DOI • title • authors • dates • deposited references

Identity, deduplication, metadata enrichment

ABSTRACT + CITATION INDEX

Scopus • Web of Science

fielded records • abstracts • citation links

Systematic discovery and citation analysis

DOMAIN DATABASE

PubMed • Europe PMC • IEEE

Xplore
subject indexing • controlled terms • domain filters

Field-specific retrieval

SCHOLARLY GRAPH

OpenAlex • Semantic Scholar

works • authors • topics • citation network

Related-paper expansion and graph traversal

PUBLISHER API

ScienceDirect • Springer

Nature
metadata and sometimes XML / HTML / full text

Document acquisition under access terms

OPEN REPOSITORY

PMC • arXiv • institutional

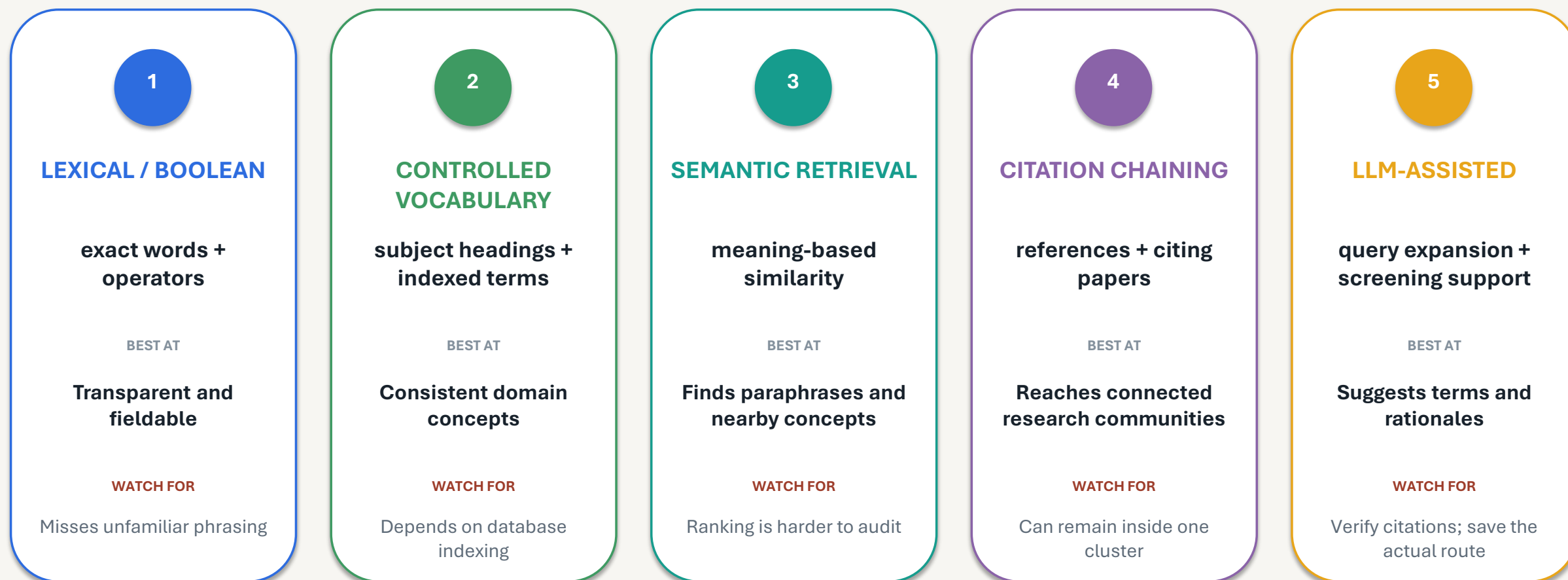
repositories
open full text • preprints • deposited manuscripts

Machine-readable or downloadable content

Search index ≠ full text. Coverage, query syntax, rate limits, subscriptions, and reuse rights differ by source.

Where you search and how you search are separate decisions

Combine retrieval methods across more than one source when the question requires broad coverage.

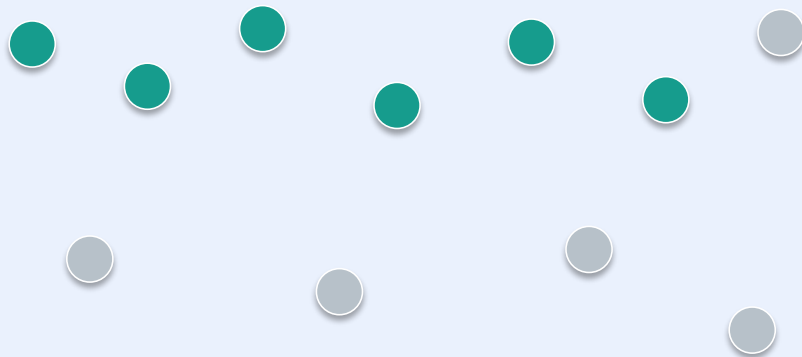


A reproducible workflow records both the source that was searched and the retrieval method used.

Retrieval quality is a precision–recall tradeoff

Discovery and screening solve different sides of the problem.

HIGH RECALL



Miss fewer relevant papers

More irrelevant results to screen

HIGH PRECISION



Most retrieved papers are relevant

Alternate terminology may be missed

Early discovery often favors recall. Screening is how you recover precision.

Screening is a transparent decision process

Predefine what counts, record why papers are excluded, and remove duplicates.

ILLUSTRATIVE FUNNEL



INCLUSION / EXCLUSION CRITERIA

- material system
- intervention or process
- outcome definition
- study type / date / language limits

WHAT TO RECORD

- DOI or stable identifier
- decision + exclusion reason
- search source and query
- reviewer / date / full-text status

Record the search itself

A paper set is more defensible when another person can reconstruct how it was found.

ILLUSTRATIVE LOG

DATABASE / SOURCE	DATE	EXACT QUERY	HITS	NOTES
Web of Science	2026-09-01	(perovskite*) AND (additive* OR passivat*) AND (stability OR degrad*)	812	broad discovery
Scopus	2026-09-01	same concept logic; database-specific syntax	765	deduplicate by DOI
Citation search	2026-09-02	forward + backward from three diverse seed papers	146	captures connected work

Save the query you actually ran — not the cleaner version you remember later.

How would you improve this weak search?

What would you add before trusting the result set?

perovskite stability

1

Which concept blocks are missing?

2

Which synonyms or alternate terms matter?

3

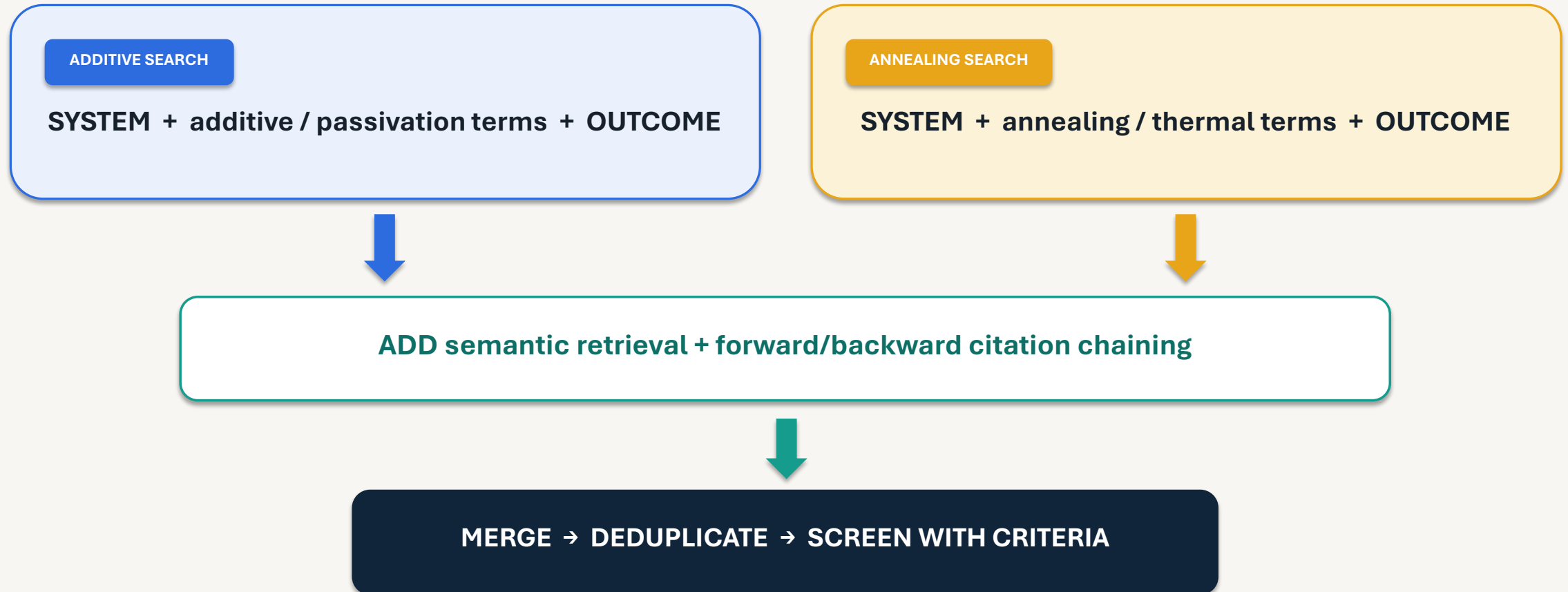
Would you use Boolean, semantic, or citation search?

4

What would count as a relevant paper?

Improve the strategy, not just the wording

There is no single perfect query; use multiple routes and test what each one contributes.



Check whether the combined strategy recovers known relevant papers without depending only on them.

Common retrieval failures — and practical controls

FAILURE

Search is too narrow

Search is too broad

Terminology changes across years

One database misses part of the field

Seed papers bias the search cluster

CONTROL

Expand synonyms; add semantic search and citation chaining

Add a missing concept block or use field filters

Iterate vocabulary using old and new seed papers

Search multiple sources and deduplicate

Use diverse seeds and independent query routes

What mining for papers should deliver

A documented evidence set — not yet a scientific dataset.



Transition: we now have the right papers — but the information is still trapped inside them.

Mining information from papers

Convert included papers into structured information without losing scientific context.



A paper must first become machine-readable

Document acquisition and parsing come before NLP or LLM-based field extraction.



PUBLISHER XML / HTML

Use native sections, references, and tagged content when available.

Most structured starting point



BORN-DIGITAL PDF

Extract the text layer and coordinates; repair columns and reading order.

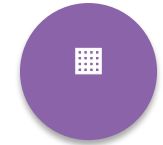
Fast, but layout can scramble text



SCANNED / IMAGE PDF

Run OCR to recover characters before any language analysis.

OCR errors can alter symbols and units



TABLES • FIGURES • SUPPLEMENTS

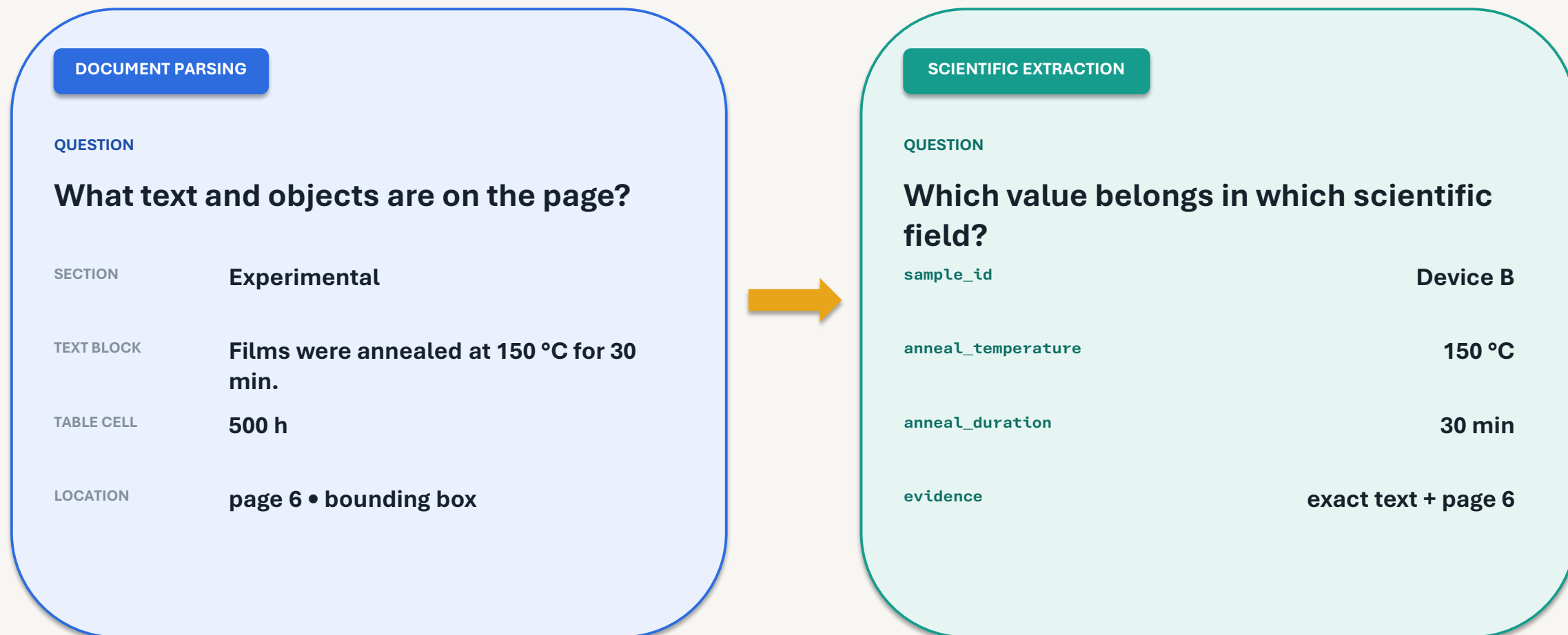
Use layout-aware table extraction, captions, vision, and separate files.

Often holds the experimental detail

OCR recovers characters. Layout parsing restores structure. Neither step has yet identified the scientific fields you need.

Text extraction is not scientific information extraction

The first recovers document content. The second maps that content into target fields and relationships.



A parser can succeed while the scientific extraction is wrong—and vice versa.

The unit of analysis is often the sample — not the paper

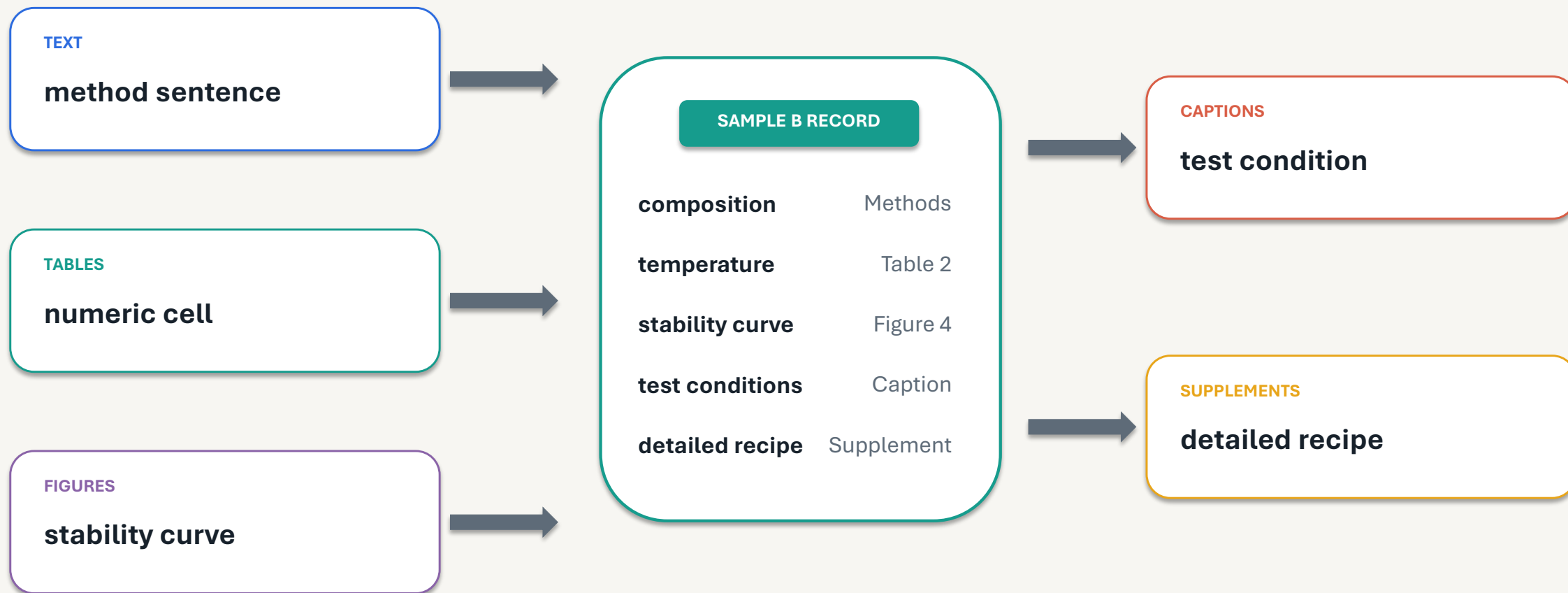
One paper can report several devices, compositions, or experiments.



Paper-level retrieval → sample-level extraction

The same sample can be described across the entire document

Sample identity must survive every jump between text, tables, figures, captions, and supplements.



Define the schema before asking a model to extract

A schema states what matters, what each field means, and how uncertainty is represented.

SCIENTIFIC QUESTION

Which processing choices are associated with operational stability?

The schema converts that question into fields that can be compared across samples.

FIELD	MEANING	EXPECTED FORM
sample_id	which device or experiment	text / local ID
process	what was done	controlled term
condition	temperature, time, atmosphere	value + unit
measurement	what was measured	defined endpoint
outcome	reported result	value + context
evidence	what supports the field	text + location

The schema is a scientific design decision, not merely an output format.

Several methods can identify target scientific fields

They are usually stacked rather than treated as mutually exclusive alternatives.

PATTERNS / REGEX

DOIs • units • dates •
stable phrases

STRONG WHEN

Fast and transparent

LIMIT

Brittle to wording and
layout

DICTIONARIES / ONTOLOGIES

materials • processes
• controlled concepts

STRONG WHEN

Consistent terminology

LIMIT

Misses unseen terms or
nuance

CLASSICAL NLP

tokenization • NER •
relation extraction

STRONG WHEN

**Entities and local
relationships**

LIMIT

Needs domain adaptation

SUPERVISED TRANSFORMERS

learn a stable labeled
task

STRONG WHEN

**Strong repeatability at
scale**

LIMIT

Requires representative
labels

LLM SCHEMA EXTRACTION

interpret context and
populate requested
fields

STRONG WHEN

**Flexible across
phrasing and schemas**

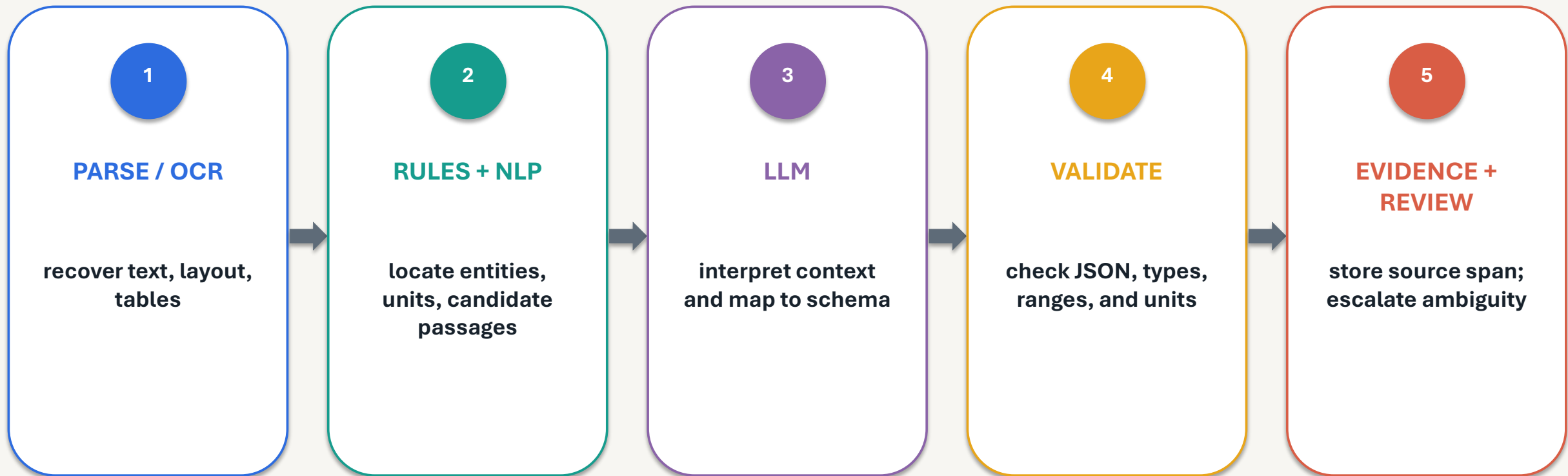
LIMIT

Can produce unsupported
values

The extraction method should match the field, document quality, available labels, scale, and acceptable risk.

OCR, NLP, and LLMs solve different parts of the pipeline

A strong system routes each task to the method that is best suited to it.



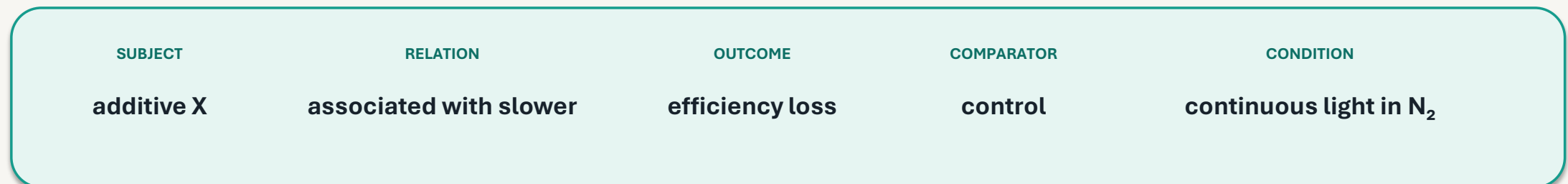
Not every document needs every stage. Use native text when available; use OCR only when the text layer is missing or unusable.

Scientific meaning lives in relationships, conditions, and uncertainty

A list of entities is useful for indexing; it is not yet a scientific claim.

“Under continuous illumination in N₂, additive X was associated with slower efficiency loss than the control.”

ENTITIES ONLY



“Associated with” preserves uncertainty; it does not silently become “caused.”

What belongs in the structured record?

Extract the fields — do not summarize.

“Device B contained 5 mol% additive X, was annealed at 150 °C for 30 min, and retained 90% of its initial power conversion efficiency after 500 h under continuous illumination in N₂.”

Write field names, not a prose summary.

SAMPLE

Which device?

PROCESS

What was done?

CONDITION

At what value and unit?

OUTCOME

What was retained, for how long, and under what test?

EVIDENCE

Where would you point a reviewer?

A reusable record preserves values, units, context, and evidence

This is an illustrative example, not a project result.

FIELD	VALUE	CONTEXT / EVIDENCE
sample_id	Device B	same device throughout
additive	X • 5 mol%	reported composition
annealing	150 °C • 30 min	process condition
stability outcome	90% initial PCE after 500 h	endpoint + duration
test conditions	continuous illumination • N₂	measurement context
evidence	exact sentence	page / section / table / figure location

PCE = power conversion efficiency

Evidence linkage is a chain of custody

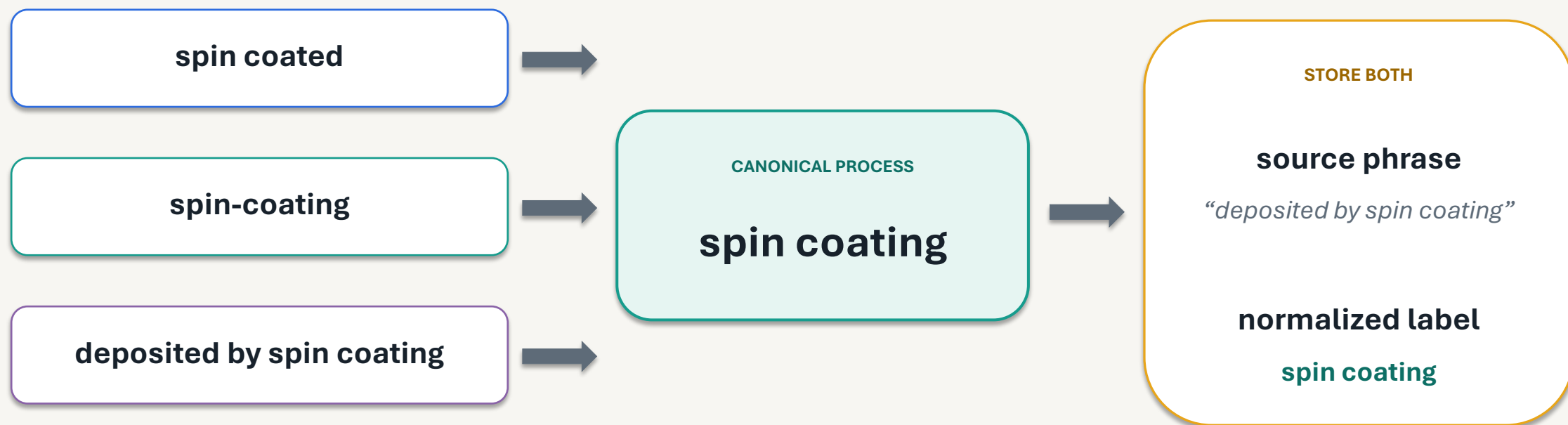
A topically related excerpt is not enough; it must support the same value and the same sample.



WHERE? page • section • table • figure **WHAT?** exact text or source object **WHICH?** paper + sample + field

Normalize terminology without erasing the source wording

Comparability and auditability require both the canonical label and the original phrase.



Normalization can include terminology, units, identifiers, and categories — but it should never hide uncertainty.

Missingness has scientifically different meanings

A blank field should not silently collapse several possible states.



REPORTED

150 °C

A value appears in the source



EXPLICITLY ABSENT

No annealing

The paper states it was not performed



NOT REPORTED / UNCLEAR

Unknown

Do not guess or silently fill

Preserving why a value is missing is often as important as preserving the value itself.

Keep three epistemic levels separate

Representation is not proof. Cross-paper analysis is not automatically causality.

1

REPORTED IN THE PAPER

“Additive X reduced defect density.”



2

STORED BY EXTRACTION

subject = X • relation = reported to reduce • object = defect density



3

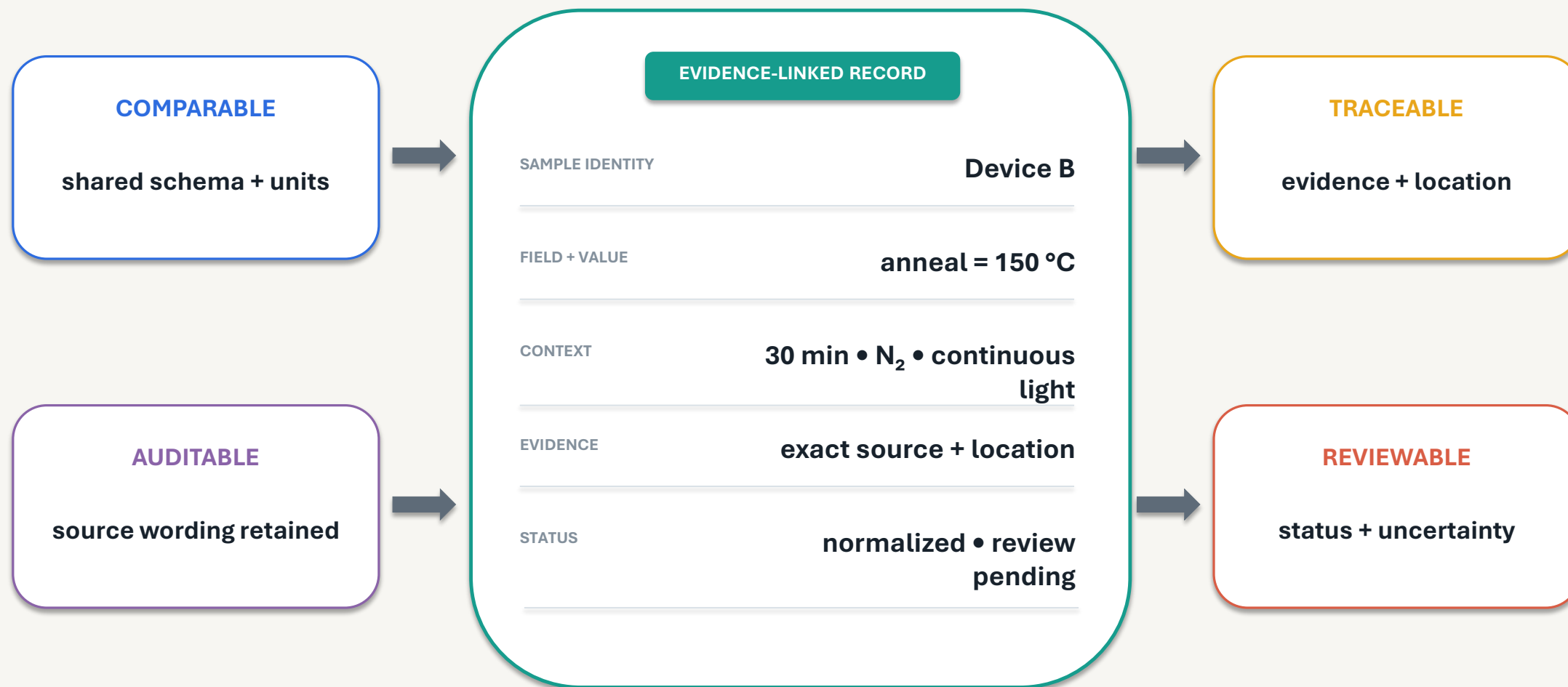
DOWNSTREAM INTERPRETATION

candidate mechanism or cross-paper hypothesis

Preserve author wording and label model-generated synthesis as interpretation.

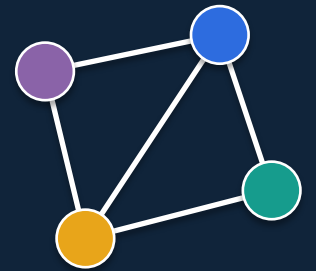
What information extraction should deliver

An answer is not enough; the record needs scientific context and a reviewable source.



Put the two jobs together

A literature-mining system is only as strong as the handoff from discovery to evidence-linked records.



A general literature-mining system is modular

Sources, parsers, extraction models, and validation can be replaced without changing the scientific handoffs.

DISCOVER + DEFINE THE PAPER SET

Sources / APIs

Search + rank

Deduplicate

Screen

Included set

ACQUIRE + UNDERSTAND THE DOCUMENT

XML / HTML / PDF

NATIVE TEXT

OCR FALLBACK

LAYOUT + TABLES

DOCUMENT MODEL

EXTRACT + VERIFY SCIENTIFIC INFORMATION

Schema

Rules / NLP / LLM

Normalize

Validate + evidence

Structured records

The architecture is generic; the search strategy, schema, ontology, and validation rules are domain-specific.

One implementation: the developed literature agent

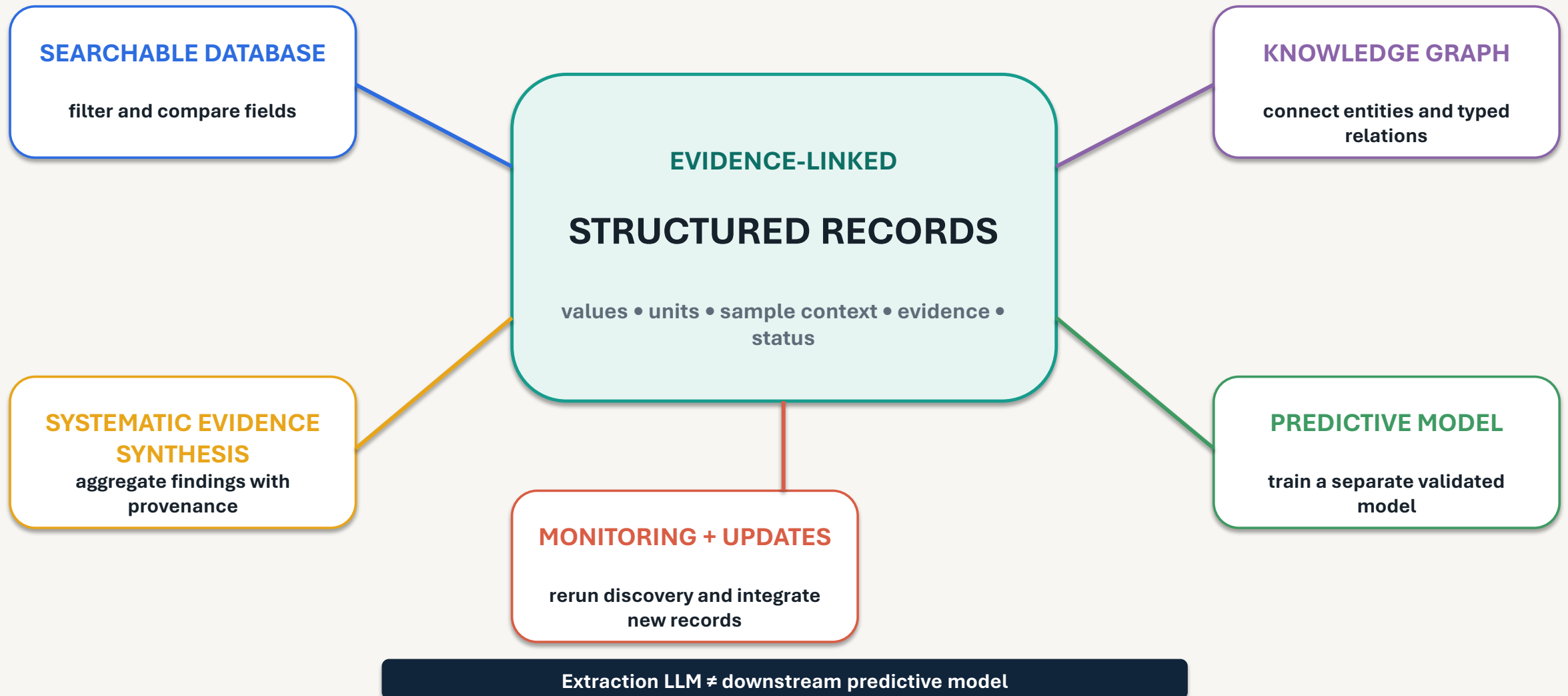
A materials-science case study of the general architecture—not the definition of literature mining.

GENERAL STAGE	CASE-STUDY IMPLEMENTATION	TRANSFERABLE DESIGN RULE
Paper discovery	curated PDFs + user-defined web search + DOI registry	make the paper set explicit
Document parsing	native text, OCR fallback, tables, captions, selected figures	preserve document structure and source location
Target extraction	schema-guided perovskite processing, performance, and stability fields	define fields from the scientific question
Quality control	JSON repair, deterministic checks, sample identity, field evidence	separate parsing, extraction, and validation
Outputs	paper bundles, integrated database, evidence packets, graphs, modeling	downstream products begin at verified records

The perovskite schema is specific. The modular handoffs and provenance requirements are general.

Verified records can support many downstream products

Every branch begins with the same evidence-linked structured record boundary.



Evaluate each layer separately — then test the full pipeline

A final field can be wrong because the paper was missed, the document was parsed badly, or the science was extracted incorrectly.

PAPER DISCOVERY

Recall	Were relevant studies found?
Precision	How much irrelevant screening burden?
Coverage	Did multiple sources and vocabularies contribute?
Reproducibility	Can the query and decisions be reconstructed?

DOCUMENT PARSING

Text completeness	Were sections and symbols recovered?
Reading order	Were columns and captions assembled correctly?
OCR quality	Were characters, units, and equations recognized?
Object recovery	Were tables, figures, and supplements captured?

SCIENTIFIC EXTRACTION

Field correctness	Are values, units, and relations correct?
Evidence support	Does the source support the same field?
Sample mapping	Was the value attached to the right experiment?
Normalization + missingness	Were transformations and unknowns preserved?

End-to-end quality fails when any layer breaks—even when the remaining components work exactly as designed.

From a research question to reusable scientific knowledge

The complete logic is broader than “send PDFs to an LLM.”



- The source determines what metadata and content are available.
- Search finds documents; parsing recovers content; extraction creates scientific fields.
- OCR, NLP, and LLMs operate at different layers and often work together.
- Evidence, validation, and human review make the output scientifically reusable.

Questions + discussion

Where could structured literature data change your research workflow?

1

Which literature task would save you the most time?

2

What field or relationship would be most valuable to extract?

3

What error would be most costly in your domain?

4

What evidence would you require before using the result?

What would you build first?